Jiefeng Zhou

Tampa, US / (+1)-779-707-0666 / jiefeng@usf.edu

Google Scholar | github.com/JeffeyChou | ORCID

Research Summary

My research makes collective communication cheaper for large-scale AI training and HPC by co-designing lossy compression with collective algorithms and GPU kernels. With collaborators at Argonne National Laboratory and NVIDIA, I built homomorphic MPI collectives on 3,584 CPU cores (SC '26), single-kernel GPU compression (PPoPP '27), and compressed Allreduce on 256 A100 GPUs. I am now building loss-aware compressed collectives for large-model training on the NCCL Device API, exploring how a collective can choose its compression format and path from measured effects on loss and time.

Education

University of South Florida (USF), Tampa, FL, U.S. 08/2025–12/2030 (expected)
Ph.D. Student in *Computer Science and Engineering*
Bellini College of Artificial Intelligence, Cybersecurity and Computing | GPA: 3.83/4.00
Advisor: Prof. Jiajun Huang
Research interests: HPC and systems for AI, collective communication, lossy compression, GPU computing

University of Electronic Science and Technology of China (UESTC), Chengdu, China 09/2021–06/2025
B.S. in *Mathematics-Physics Fundamental Science*
Yingcai Honors College | GPA: 3.88/4.00
Research advisor: Prof. Yong Deng
Research interests: evidence theory, information entropy

Publications

Conference Papers

- **Zhou, J.**, Di, S., Cappello, F., & Huang, J. (accepted). MAZ: Advancing the ratio–throughput frontier with multi-dimensional adaptive compression. In *Proceedings of the ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming (PPoPP '27)*. Association for Computing Machinery.
- **Zhou, J.**, Di, S., Guo, Y., Guo, Y., Zhang, L., Zhao, K., Thakur, R., Cappello, F., & Huang, J. (accepted). hZCCL2: Co-designing collective communication with vectorized homomorphic compression. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC '26)*. IEEE.

Journal Articles

- **Zhou, J.**, Li, Z., Cheong, K. H., & Deng, Y. (2025). Limit of the maximum random permutation set entropy. *Physica A: Statistical Mechanics and its Applications*, *664*, 130425.
- Zhan, T., **Zhou, J.**, Li, Z., & Deng, Y. (2024). Generalized information entropy and generalized information dimension. *Chaos, Solitons & Fractals*, *184*, 114976.
- **Zhou, J.**, Li, Z., & Deng, Y. (2024). An improved information volume of mass function based on plausibility transformation method. *Expert Systems with Applications*, *237*, 121663.
- **Zhou, J.**, Li, Z., & Deng, Y. (2024). Random walk in random permutation set theory. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, *34*(9), 093137.

Posters

- **Zhou, J.** (2026). *HV-Coll: Co-designing collective communication with vectorized homomorphic compression*. ACM Student Research Competition (SRC) Poster, SC '26, Chicago, IL, USA.

In Preparation

- **Zhou, J.**, Di, S., Guo, Y., Guo, Y., Zhang, L., Zhao, K., Tu, Y., Thakur, R., Cappello, F., & Huang, J. (in preparation). *ghZCCL2: High-throughput homomorphic compression with topology-aware co-design for GPU collectives*.

Selected Research Projects

hZCCL2: Collective Communication with Vectorized Homomorphic Compression SC '26

- Co-designed an AVX-512 vectorized error-bounded compressor with outlier-aware encoding and homomorphic reduction (up to 5.31× faster compression than fZ-light; 8.1× faster than decompress–operate–compress).
- Achieved up to 43.8× Allreduce and 67.6× Alltoall speedups over native MPI on 3,584 Intel Xeon Max cores, and a 39.5× speedup in an image-stacking application.

ghZCCL2: Topology-Aware Homomorphic Compression for GPU Collectives In preparation

- Built a high-throughput GPU homomorphic compressor, an Allreduce that stays compressed through every step, and a topology-aware Allreduce pairing raw NVLink transfers with compressed inter-node reductions.
- Compressor 1.49× faster than the best homomorphic baseline; on 256 A100 GPUs, geometric-mean Allreduce speedups of 5.30× over Cray MPI, 2.80× over NCCL, and 2.06× over the best homomorphic collective.

MAZ: Multi-Dimensional Adaptive GPU Compression PPoPP '27

- Designed the first single-kernel GPU compressor with adaptive multi-order 2D/3D prediction, via a register-carried stencil state, a single-read predictor-search lattice, and a wavefront-free decoder.
- On an NVIDIA B200 across 79 scientific fields, improves the compression ratio of FSZ by 1.69× at comparable throughput and of cuSZp3 by 1.61×, at about 5× the throughput of the strongest-ratio baseline.

Compressed GPU-Initiated Collectives for Large-Model Training Ongoing

- Profiled Megatron-LM training of OLMoE-1B-7B on 32 A100 GPUs (ALCF Polaris): the expert-parallel Alltoall alone took 37% of each step, with almost no communication hidden behind computation.
- Building collectives on the NCCL Device API with a 0.9–1 TB/s compressor on A100: step time 16.7% shorter than NCCL on OLMoE-1B-7B (16 nodes) and about 9% shorter than COCCL on a dense 7B model, at matched loss.

Industry & Research Experience

High Performance & Intelligence Lab (π Lab), Tampa, United States 08/2025–Present

Research Assistant | University of South Florida | Advisor: Prof. Jiajun Huang

- First-authored hZCCL2 (SC '26) and MAZ (PPoPP '27) on compressed collectives and GPU lossy compression.
- Collaboration with Sheng Di's group at Argonne and with Yanfei Guo (NVIDIA).

Tang Lab, Chengdu, China 12/2024–07/2025

Research Intern | Peking University Chengdu Academy for Advanced Interdisciplinary Biotechnologies

- Developed a language model for single-cell perturbation prediction as my undergraduate thesis project.

Information Fusion and Intelligent Systems Lab, Chengdu, China 02/2023–09/2024

Undergraduate Research Assistant | University of Electronic Science and Technology of China

- Under Prof. Yong Deng, published four journal papers on evidence theory and information entropy (first author on three), in *Physica A*, *Chaos*, *Chaos, Solitons & Fractals*, and *Expert Systems with Applications*.

Awards & Honors

- **Travel Grant** | SC26 ACM Student Research Competition 09/2026
- National Grant ×3; Dream Scholarship, China Education Development Foundation (National Level) 2022–2024
- Model Student Scholarship ×2; Outstanding Undergraduate Scholarship (University Level) 2022–2024
- **Top 7%** | The 17th IEEEXtreme Programming Competition (Global Level) 12/2023
- **First Prize** | The 23rd Mathematical Modeling Competition of UESTC (University Level, **Top 5%**) 05/2023

Professional Service

Junior Program Committee @ ACM International Conference on Supercomputing (*ICS 2027*) 12/2026

Peer Reviewer @ *Expert Systems with Applications* 11/2023